Hardware Interfacing & Circuit Schematic Diagram

Comprehensive Pin Connection, Signal Routing, and Power Subsystem Architecture

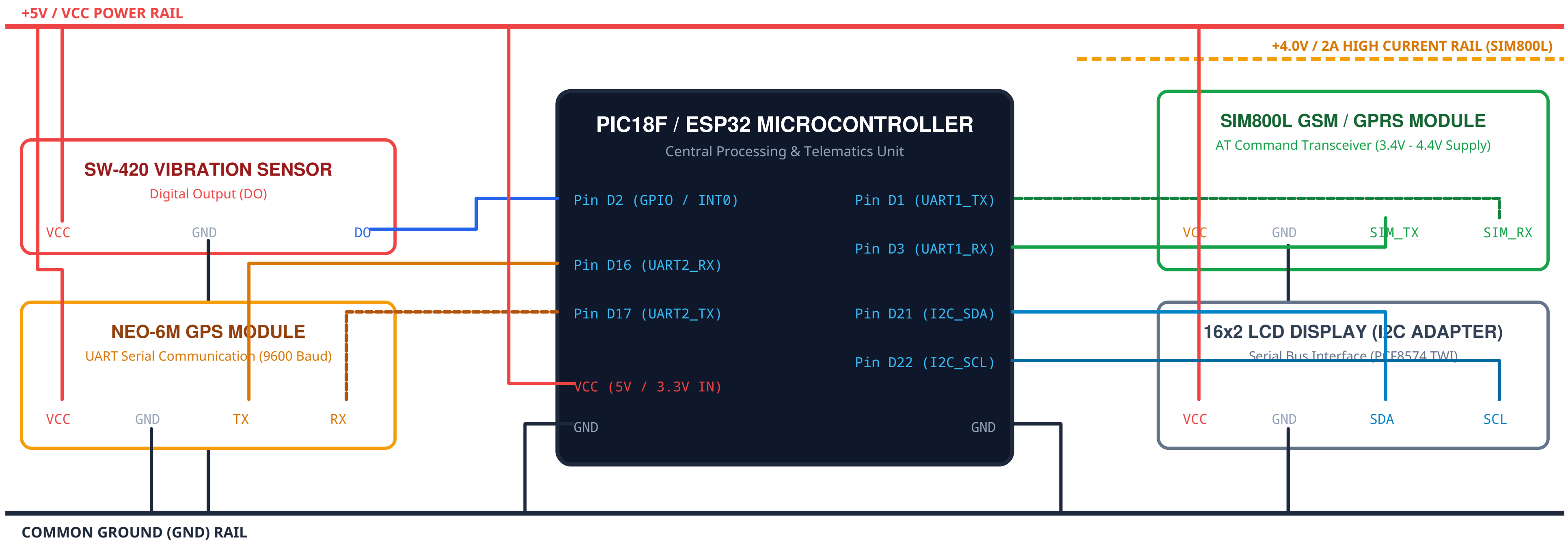

FIGURE: BROAD SCHEMATIC INTERCONNECT DIAGRAM FOR ACCIDENT DETECTION & ALERT HARDWARE PLATFORM

Component Interconnection Table

Module Name	Module Pin	Target Microcontroller Pin	Signal Type / Operating Voltage	Functional Description
SW-420 Sensor	VCC	+5V / +3.3V Power Rail	Power Input (3.3V - 5V DC)	System power supply for piezoelectric comparator
	GND	Common Ground Rail	Ground Reference (0V)	Ground return path
	DO (Digital Out)	GPIO / INT0 Pin (Pin D2)	Digital Input (Active High/Low)	Sends interrupt signal upon exceeding shock threshold
NEO-6M GPS	VCC	+5V / +3.3V Power Rail	Power Input (3.3V - 5V DC)	Powers GPS baseband receiver engine
	GND	Common Ground Rail	Ground Reference (0V)	Ground return path

Module Name	Module Pin	Target Microcontroller Pin	Signal Type / Operating Voltage	Functional Description
	TX	UART2_RX (Pin D16)	Serial Data Input (3.3V TTL)	Transmits NMEA location strings ($GPRMC) to MCU
	RX	UART2_TX (Pin D17)	Serial Data Output (3.3V TTL)	Receives configuration command frames from MCU
SIM800L GSM	VCC	Dedicated Buck Converter (+4.0V)	Power Supply (3.4V - 4.4V / 2A Peak)	High current supply rail to sustain cellular transmission bursts
	GND	Common Ground Rail	Ground Reference (0V)	Common ground reference for serial communication
	TX	UART1_RX (Pin D3)	Serial Data Input (3.3V TTL)	Transmits modem responses and AT return codes to MCU
	RX	UART1_TX (Pin D1)	Serial Data Output (3.3V TTL)	Receives AT message sending commands (`AT+CMGS`) from MCU
16x2 LCD (I2C)	VCC	+5V Power Rail	Power Input (5V DC)	Display backlight & logic power supply
	GND	Common Ground Rail	Ground Reference (0V)	Ground return path
	SDA	I2C_SDA (Pin D21)	Bidirectional Serial Data	Serial Data bus line for PCF8574 backpack adapter
	SCL	I2C_SCL (Pin D22)	Serial Clock Input	Serial Clock synchronization bus line

Hardware Assembly & Power Regulation Notes:
1. **GSM Power Supply:** Never power the SIM800L module directly from the 5V microcontroller pin. It requires up to 2A during network registration spikes; use an LM2596 Buck converter calibrated to 4.0V.
2. **Common Ground:** Ensure all module grounds (SW-420, GPS, GSM, LCD, and MCU) are connected together to form a common ground plane.